

# AYUSH GHOSH

U.S. Citizen

☎ 443-917-8321

✉ [ayushrghosh@berkeley.edu](mailto:ayushrghosh@berkeley.edu)

🌐 [linkedin.com/in/ayush-ghosh](https://www.linkedin.com/in/ayush-ghosh)

🌐 [Portfolio](#)

## Education

### University of California, Berkeley

Expected May 2028

*Mechanical Engineering, B.S.; Minor in Electrical Engineering and Computer Science*

GPA: 3.95

- **Relevant Coursework:** Engineering Design and Development, Computer-Integrated Manufacturing, Three-Dimensional Modeling and Design, Mechanics, Thermodynamics, Introduction to Circuits and Devices, Electricity and Magnetism

## Experience

### Volchor

September 2025 – May 2026

*Mechanical Engineering Intern – Signed NDA*

*San Francisco, CA*

- Analyzed and qualified **structural components** in **ANSYS** under combined aerodynamic and thermal loading, applying **factors of safety** and material allowables for a high-acceleration flight environment
- Wrote **MATLAB** physics models of payload dynamics (stability, transient loads, kinematics) to inform structural sizing and detachable heat-flux-absorbing shell concepts
- Modeled vehicle geometry and supporting **vacuum infrastructure** in **SolidWorks**, including pressure boundaries, isolation gates, and pumps

### FSAE Formula Electric, UC Berkeley

January 2026 – Present

*Accumulator Mechanical Engineer*

*Berkeley, CA*

- Co-designed the team's custom **140S3P** lithium-ion high-voltage battery pack from concept through fabrication, driving **GD&T** tolerance stack-ups, **materials selection** for dielectric insulation, **thermal management**, and **design for manufacturability and assembly (DFMA)**
- Instrumented a **validation test fixture** with an **Arduino-based thermocouple array**, **four-point resistance** measurement, and **Python** data logging to verify **ANSYS** thermal simulations, root-causing hotspots that drove busbar redesigns and cut accumulator mass by **18%**
- Developed a waterproof, electrically insulated **HV/LV enclosure** with a **silicone gasket** seal for **environmental protection**, and packaged **cell-monitoring PCBs** into the module housing alongside the **electrical team**

### KwikKart

May 2026 – August 2026

*Hardware Engineering Intern*

*Austin, TX*

- Re-engineered a smart tablet's rear **structural assembly** in **SolidWorks**, packaging electronics, cameras, and a spring-loaded, tool-free **battery ejection mechanism** while cutting assembly mass by **30%**
- **Prototyped** and validated **forced-air thermal management** for an **NVIDIA Jetson AGX Orin**, iterating a **CNC-machined** heatsink and chimney flow path in **ANSYS CFD** to meet tight packaging constraints
- Authored **KiCad** schematics of the tablet's full electronic architecture, streamlining **mechanical–electrical integration** and debugging

## Projects

### Custom Quadcopter

June 2026 – August 2026

*Personal Project*

- Sized motors, propellers, ESCs, and battery from first principles and validated a **~0.9 kg SolidWorks** airframe with **FEA**, achieving a **~3.6:1** thrust-to-weight ratio and **~12 min** flight time
- Fabricated a tethered **flight-test fixture** and tuned **PID** gains to a stable hover within **±2°** roll/pitch under disturbances
- Laid out a **6-layer STM32F405 KiCad** flight-controller PCB (IMU, barometer, RC receiver) and wrote firmware from scratch for sensor fusion, attitude stabilization, and motor control

### Computed Axial Lithography 3D Printer

July 2026 – Present

*Design for Nanomanufacturing Lab, UC Berkeley*

- Building HeliCAL60, a benchtop **computed axial lithography (CAL)** 3D printer that cures full parts volumetrically by projecting tomographic light patterns into a rotating vial of photopolymer resin
- Designing the printer's **positioning mechanisms** in **SolidWorks**, including mounts and structural interfaces for **5 μm-resolution** motorized translation stages and a **lead-screw drive** for repeatable axial positioning
- Implementing a **closed-loop alignment system** that pairs a **CMOS** image sensor with the motorized stage, using real-time feedback to auto-correct optical/rotational misalignment

## Skills

**CAD (MCAD/ECAD):** CATIA V5, Siemens NX, SolidWorks, PTC Creo, Fusion 360, Onshape, Altium Designer, KiCad, GD&T

**Mechanical Design:** Structures, Mechanisms, Thermal Management, Electronics Packaging, Materials Selection, DFM/DFMA

**Analysis:** ANSYS (Mechanical, Thermal, Fluent CFD), FEA, Structural & Thermal Analysis, Verification & Validation

**Test & Fabrication:** Test Fixture Design, Hardware Prototyping, CNC Machining, CAM, Sheet Metal, 3D Printing

**Electrical:** HV Battery Pack Design, HV/LV Insulation, PCB Design & Assembly, Soldering

**Programming:** C/C++, Python, MATLAB, Java, Bare-Metal Firmware